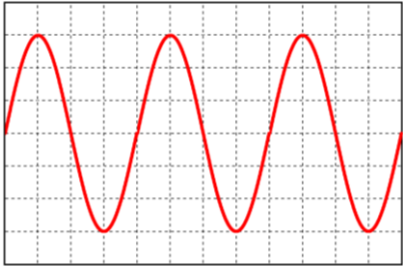


|               |                                                                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>4</b>      |                                                                                                                                                                                                   |
| <b>(a)(i)</b> | When the voltmeter reading is zero, the p.d. across AB is equal to the p.d. across the $100\ \Omega$ resistor.                                                                                    |
|               | $\frac{x}{l_{AC}} = \frac{100}{100 + 160}$ $x = \left( \frac{100}{100 + 160} \right) (90 \times 10^{-2})$ $= 34.6 \times 10^{-2}\ \text{m}$ $\approx 35\ \text{cm}$                               |
| <b>(ii)</b>   | $R_{\text{eff}} = \frac{\varepsilon}{I}$ $= \frac{3.0}{0.234}$ $= 12.82\ \Omega$                                                                                                                  |
|               | $\frac{1}{R_{\text{eff}}} = \frac{1}{R_{AC}} + \frac{1}{R + 160}$ $\frac{1}{12.82} = \frac{1}{R_{AC}} + \frac{1}{100 + 160}$ $R_{AC} = 13.49$ $\approx 13.5\ \Omega$                              |
| <b>(iii)</b>  | If the battery has internal resistance, the terminal potential difference across the battery will be smaller. With the same current, the effective resistance of the external circuit is smaller. |
|               | Hence, the resistance of the resistance wire AC will be a smaller value.                                                                                                                          |

|               |                                                                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(b)(i)</b> | <p>For half wave rectification,</p> $V_{\text{rms}} = \sqrt{\frac{V_0^2 \left(\frac{T}{2}\right)}{T}}$ $= \frac{V_0}{2}$                                                              |
|               | $P_{\text{ave}} = \frac{V_{\text{rms}}^2}{R}$ $= \left(\frac{V_0}{2}\right)^2 \left(\frac{1}{R}\right)$ $= \left(\frac{12}{2}\right)^2 \left(\frac{1}{13.5}\right)$ $= 2.7 \text{ W}$ |
| <b>(ii)</b>   | <p>Sinusoidal waveform with an amplitude of three divisions since the peak voltage is 12 V.</p>                                                                                       |
|               | $T = \frac{1}{f}$ $= \frac{1}{50}$ $= 0.020 \text{ s}$ $= 20 \text{ ms}$ <p>Period of each cycle is 4 divisions and three cycles are drawn.</p>                                       |
|               |                                                                                                    |

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